Supplementary Methods: object-switch trajectory analysis and human-AI workflow

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1. Empirical-object hierarchy

The study separates three levels that are easily conflated.

1. **Implementation objects:** model weights, attention heads, MLPs, residual blocks and layers. These are the engineered objects that realize inference.
2. **Empirical process object:** the ordered hidden-state sequence $T_{m,p} = (h_{m,p,1}, \ldots, h_{m,p,L_m})$. This is the object on which the present mechanism comparisons are defined.
3. **Coordinates and diagnostics:** direct hidden summaries, hidden-state flow, vocabulary-facing TopK/VIM readbacks, the decision-window clean-relative margin profile, ΔU, transition coordinates and residual diagnostics.

The methodological statement "empirical research object before coordinates" means choosing a testable process object before selecting measurements. It does not claim that engineering objects are unimportant or make a metaphysical claim about the layer-state trajectory.

H-shape denotes the ordered profile of clean-relative candidate margins across a specified layer window. It is a trajectory-shape summary, not a hidden-state tensor or an independent mechanism label. Earlier analyses also used related shorthand for window summaries and previous-transition history; those objects are kept distinct here. The decision-window H-shape is exactly dR over decision layers and is construction-aligned with ΔU. Legacy labels remain only in the Source Data manifest for provenance.

2. Intellectual provenance and operational boundaries

Manuscript object	Mathematical or physical source idea	Operational use here	Identity explicitly not claimed
Layer-state trajectory	State trajectory of a discrete dynamical system	Ordered sequence of last-token hidden states across blocks	Layer depth is not physical time
Exploratory registration	Functional or phase registration	Data-selected windows provide model-specific indexing choices for downstream analyses	Prospective joint-role recurrence failed; windows are not confirmed phases or identical coordinates
ΔU	Reaction coordinate and order-parameter compression	Oriented PC1 of a clean-relative decision-window margin matrix	Not energy, potential, thermodynamic order parameter or full mechanism
Transition closure	State-space and reduced-order closure	Test whether current-state and realized-transition coordinates compactly describe the observed update	Not an autonomous dynamical law when target-transition information enters
Chord-directed geometry	Finite-path alignment, detour and turning	Compare real order with endpoint-preserving shuffles under a declared cosine geometry	Not a learned metric, exact geodesic, least action or conservation law
Structured residual	Residual diagnostics in reduced-order modelling	Test whether deviation from the chord-directed reference retains information	Not a complete tangent-normal decomposition
Local-transition actuator	Feedback-control distinction between state estimation, abstention and actuator choice	Mechanism-conditioned transition operator plus confidence-gated bounded action	Crosses a declared emitted-token boundary in three checkpoints; not answer-correctness, safety or deployment control

The source ideas were introduced after empirical problems required a compact description. They were not used as labels that made an observation true. Negative results narrowed each analogy.

3. Controlled prompt subset

The direct hidden-state analyses used eight complete graph groups selected with seed 20260610: group IDs 0, 2, 15, 23, 25, 26, 51 and 63. All 12 conditions per group were retained, giving 96 prompts. The same source rows were used for all three checkpoints. This design preserves within-group condition structure while making 50 layer-order shuffles tractable. It is not a first-row convenience sample.

The subset manifest contains prompt ID, group and graph IDs, raw condition, analysis label, task-defined clean and conflict candidates, source row and model-specific readback records. Final generated-answer correctness is not used in ΔU, coordinate-audit labels, intervention-class labels or policy labels.

Each prompt encodes a short relation chain from named entities to one of two declared answer candidates. The clean candidate is supported by the unperturbed chain. The conflict candidate is introduced by a distractor, competing statement, update or exception. Stable conditions retain a decisive clean chain, competition conditions support both candidates, and closure conditions redirect the chain. These labels are fixed by prompt construction before model fitting. They are analysis classes, not dynamical states discovered from the hidden trajectories. The task was selected to isolate evidence competition and updating under controlled counterfactual conditions; no claim is made that it represents open-ended reasoning.

Supplementary Methods Table 1 | Analysis-label and information-boundary crosswalk.

Analysis family	Source object and raw conditions	Fitted target or endpoint	Inputs permitted	Information explicitly excluded
Coordinate audit and 96-prompt geometry subset	Twelve-condition panel. Stable: clean, rename, irrelevant, paraphrase and weak distractor (stored as stable-shift). Competition: balanced competition, direct conflict, equal evidence and source claim (stored as source-ambiguity). Closure: closure update, closure override and exception override.	Three-class prompt-condition label for proxy comparison or plot grouping	Prompt metadata for labels; fold-fitted predecision state and the audited proxy for prediction	ΔU, decision-window clean-relative profile, generated answer and correctness
Functional-window scan	Separate 12-condition discovery panel. Stable: clean, rename, permuted, redundant, irrelevant and paraphrase; weak distractor stored as stable-shift and collapsed with stable. Competition: balanced competition and direct conflict. Closure: closure update, closure override and exception override.	Exploratory preservation, class-information and downstream-coupling score	Candidate window measurements within graph-aware folds	Confirmatory interpretation; selection is not nested
Prospective functional-window boundary	Same 12 templates on 24 new graph groups with disjoint entities, relation phrases and candidate labels	Frozen-window role recurrence and rank	Discovery-only PCA and fitted models; confirmation labels only for final scoring	Window reselection or confirmatory claims after a failed gate
ΔU construction	Declared clean and conflict candidate identities within each graph group	Oriented PC1 score of the clean-relative decision-window matrix	Candidate margins in the declared window; training-fold PCA where transfer is evaluated	Generated answer, final correctness and an analysis-independent mechanism label
Internal-control and confidence-gated intervention	Separate ten-condition panel. Stable: clean, redundant, irrelevant and paraphrase. Competition: weak distractor, balanced competition and direct conflict. Closure: closure update, closure override and exception override.	Three-class intervention label used by the mechanism classifier and gate	Training-graph prompt labels and pre-intervention trajectory features	Held-out outcomes, generated answer and correctness
Policy supervision	Eight-action internal library or expanded ten-action boundary library	Discrete (α, β) action maximizing the declared training utility; `none` is $(0, 0)$	Training-graph intervention responses and fitted mechanism features	Held-out action outcomes and correctness
Candidate-boundary endpoint	Paired no-intervention and controlled continuations	Conflict-to-clean crossing of the two declared candidates; in the expanded test, strict full-vocabulary top-1 crossing	Held-out outputs only for evaluation	Use as a fitting label; interpretation as correctness or open-ended answer control

The coordinate-audit and intervention panels therefore do not share a universal latent class variable. In particular, weak distractor is stored as stable-shift and collapsed with stable in the coordinate audit, but is grouped with competition in the separately generated intervention panel. Both mappings were fixed in their corresponding data-generation and analysis scripts before model fitting. They are retained rather than harmonized after observing outcomes.

4. Direct hidden-state update summaries for Fig. 1

For normalized states $x_l = h_l/\|h_l\|_2$, adjacent hidden-state change is

$$c_l = 1 - \cos(x_l, x_{l+1}).$$

Let $s_l = x_{l+1} - x_l$ and hidden size d. The normalized participation ratio is

$$P_l = \frac{(\sum_{j=1}^{d} s_{l,j}^2)^2}{d\sum_{j=1}^{d} s_{l,j}^4}.$$

P_l approaches one when step energy is evenly distributed and decreases when fewer dimensions dominate. Fig. 1 reports means and standard errors by collapsed coordinate-audit condition family and checkpoint. These families are predefined plot groups, not inferred trajectory phases. The calculations use native hidden states and no vocabulary projection. Executable filenames are listed in the Source Data manifest.

5. Readback and projection-selection controls

For clean candidate token c_p, conflict candidate token e_p and output matrix W,

$$R_{p,l} = h_{p,l}^{\top}(W_{c_p} - W_{e_p}).$$

TopK/VIM features summarize projected vocabulary neighbourhoods. The row-permuted control shuffles rows of W, preserving matrix values but changing token correspondence. The Gaussian control uses a seeded random matrix of matched shape. Because these controls operate after hidden-state extraction, they audit coordinate dependence of the vocabulary-facing geometry without changing the native hidden-state trajectories.

For the independent prompt-class audit, every proxy augmented the same predecision base state. Proxy PCA was capped at 32 dimensions and fitted separately inside each training fold. Five-fold GroupKFold used relation graph as the split unit. In the 12-condition source panel, clean, rename, irrelevant and paraphrase were stored as stable; weak distractor as stable-shift; balanced competition, direct conflict and equal evidence as competition; source claim as source-ambiguity; and the three update or override conditions as closure. Stable and stable-shift then mapped to the stable classifier class; competition and source-ambiguity mapped to competition; closure mapped to closure. Macro F1 is the unweighted mean of class-specific F1 scores. The reported value is augmented minus base prompt-class macro F1. The label map is fixed by controlled prompt metadata and does not use ΔU, generated answers or the decision-window clean-relative margin profile.

Algorithm 1: independent coordinate audit

```
for each checkpoint and proxy family:
    partition relation graphs into five folds
    for each held-out fold:
        fit all preprocessing, including proxy PCA, on training graphs only
        fit the base mechanism classifier on predecision state features
        fit the augmented classifier on the same base plus the proxy
        predict the held-out graphs
    concatenate held-out predictions across folds
    report macro-F1(augmented) - macro-F1(base)
```

6. Exploratory functional-window scan and prospective boundary test

Candidate windows span the first 45% of checkpoint depth. Window lengths are 2, 3, 4, 5, 6 and 8, step size is one layer, and TopK neighbourhood sizes are 100 and 500. For each window, the preservation score is one minus the mean clean-relative distance for relation-preserving conditions. The source field was named topology, but the quantity does not compare relation-preserving with structure-changing conditions and is not itself a topology-separation test. Prompt-condition-class information is GroupKFold macro F1, and downstream coupling is a GroupKFold correlation with ΔU. The objective is

$$S = 0.5S_{\text{preservation}} + 0.8\max(S_{\text{class}}, 0) + 0.8\max(S_{\text{downstream}}, 0).$$

The maximum-scoring interval is retained. Selection is non-nested because the same scan supplies candidate and selection score. GroupKFold prevents graph overlap in component estimates but does not remove selection optimism. The windows are exploratory and data-selected.

Algorithm 2: exploratory functional registration

```
for each checkpoint:
```

```
enumerate all permitted starts, lengths and neighbourhood sizes

for each candidate window:

    calculate clean-relative preservation

    calculate group-held-out prompt-condition-class macro F1

    calculate group-held-out downstream correlation

    combine the three quantities using the stated score S

retain the maximum-scoring window

label the result exploratory because selection and evaluation share the scan
```

6.1 Prospective confirmation protocol and result

After the exploratory scan, the overall windows and neighbourhood sizes were frozen as Qwen layers 5-6 with $K = 500$, Llama layers 0-2 with $K = 500$, and Gemma layers 1-6 with $K = 100$. The prospective confirmation set comprised 24 new relation graphs and the same 12 perturbation templates. It replaced every discovery entity triple and relation pair and used 12 candidate labels drawn from an audited common one-token vocabulary that was disjoint from the discovery candidate labels. No confirmation prompt entered the original scan.

For each checkpoint, the discovery rows alone fitted the clean-relative decision-profile PCA, its sign orientation, feature scaling, a balanced three-class logistic classifier and ridge regression to ΔU. The fitted objects were then applied to confirmation rows. Confirmation labels were used only to score the predeclared endpoints. The full candidate pool was evaluated to obtain the rank of the already frozen window; no new winner replaced it. Two thousand graph-group bootstraps produced 95% intervals. Nine hundred and ninety-nine discovery-label permutations formed the class-information null, and 999 within-discovery-graph ΔU permutations formed the downstream null.

The frozen per-checkpoint gate required: (i) a positive 95% bootstrap interval for the graph-paired difference between structure-changing and relation-preserving composite distances; (ii) macro F1 at least 0.55 and greater than the label-null 95th percentile; and (iii) positive downstream correlation greater than the target-null 95th percentile. Window-location recurrence was declared separately when the frozen window ranked in the top quartile of the unchanged candidate pool. The protocol and its SHA-256 hash were written before extraction. One initial run stopped after Qwen extraction because the imported feature function returned two objects rather than three; no endpoint was calculated. The interface correction and failed-run logs are retained.

Supplementary Methods Table 2 | Prospective functional-window confirmation on 24 untouched graph groups.

Checkpoint	Topology separation (95% bootstrap interval)	Class macro F1 / label-null 95th	Downstream r / target-null 95th	Frozen rank	Joint role gate	Location gate
Qwen	-0.185 (-0.189, -0.181)	0.739 / 0.521	0.659 / 0.623	58/112	Fail	Fail
Llama	-0.092 (-0.092, -0.091)	0.662 / 0.510	0.411 / 0.452	38/40	Fail	Fail
Gemma	-0.090 (-0.091, -0.089)	0.649 / 0.475	0.624 / 0.404	5/100	Fail	Pass

The class-information endpoint transferred in all checkpoints. Downstream coupling passed its frozen null in Qwen and Gemma but not Llama. Topology separation reversed in every checkpoint, and only Gemma retained top-quartile location. The three-checkpoint role-recurrence claim is therefore rejected. These results do not invalidate use of the discovery windows as explicitly exploratory analysis choices, but they prohibit treating the windows as confirmed functional phases or homologous cross-checkpoint locations.

7. Executable ΔU definition

Within each graph group, the clean condition supplies a baseline. If a clean row is absent, the group mean is used. The clean-relative margin is

$$dR_{p,l} = R_{p,l} - R_{p_0,l}.$$

For mapped decision layers $\mathcal{L}_{\text{dec}}$, form

$$D = [dR_{p,l}]_{p,l\in\mathcal{L}_{\text{dec}}}.$$

PCA is fitted to D, and ΔU_p is its PC1 score. The sign is oriented so that stable or stable-shift rows have greater mean scores than closure rows when those groups are available. Candidate identities from the controlled task enter R; the model's generated answer and final correctness do not. ΔU is therefore a task-constructed exploratory diagnostic of declared candidate-path separation. Its numerical axis is not transferred to a different prompt family. The controlled lexical and addition tests fit new task-specific coordinates with the same protocol and report held-out separation independently.

The decision-window clean-relative margin profile is $D_{p,:}$. It cannot serve as independent evidence for ΔU because both are the same construction at different dimensionalities. Correlation with a dominance readout derived from related decision-window information is a sanity check. Independent bounding analyses remove the direct margin, exclude decision and endpoint information, change the window, test residual signal or compare intervention nulls.

8. Transition-coordinate closure

Previous-transition history features comprise the preceding clean-relative margin change, preceding rank-gap change, reversals in both signals, and backtracking relative to the preceding and initial centres. The realized transition coordinate O_l^{cont} comprises the first ten principal components of the observed transition representation. Closure models compare state-only, history-only, combined state-history, state plus O_l^{cont}, and bilinear state-transition feature sets.

If adjacent target-layer information enters O_l^{cont}, the analysis is descriptive factorization. Macro R^2 is the unweighted mean of target-wise R^2 values. The predictive audit estimates $\hat{O}_l^{\text{cont}}$ from current-state, previous-transition history and optional layer/condition context under graph-grouped cross-validation, excluding the realized target transition. Within each outer fold, the first-stage model predicts $\hat{O}_l^{\text{cont}}$ separately for training and test rows. The second-stage model then constructs state-coordinate products independently from each fold's own predictions; no observed O_l^{cont} enters that stage. State-only closure reached macro $R^2 = 0.822$, observed O_l^{cont} reached 0.979, and the best target-excluded bilinear model using $\hat{O}_l^{\text{cont}}$ reached 0.862. Its increment over state only was 0.0394 (95% prompt-group bootstrap interval, 0.0368-0.0426), and it retained 25.1% of the observed-coordinate gain. Models predicting O_l^{cont} itself ranged from macro $R^2 = 0.41$ to 0.61. All four corrected bilinear feature sets completed five graph-grouped folds and improved over their matched additive versions by 0.0279-0.0318 macro R^2. This is target-excluded incremental prediction, but remains exploratory and does not establish a task-general forward law. Metric-specific embedding diagnostics are not combined because distance and correlation scores have different scales and interpretations.

Prospective final candidate-margin forecast

This separate analysis predicts the exact final clean-minus-conflict output-logit margin from partial decision-window histories. The frozen relation-graph split contains 67 training groups (670 prompts) and 29 held-out groups (290 prompts). At cutoff layer k, the ordered-history predictor contains only the candidate-margin values observed at decision layers $l \le k$. It excludes all later states and transitions, the final generated token and answer-correctness labels. Ridge regression uses penalty 10. A balanced logistic regression with $C = 1$ predicts whether the final candidate margin is positive from the same features.

For the order null, the current-cutoff value is fixed and all earlier values are independently permuted within each prompt. This avoids the feature-renaming invariance of a common permutation followed by model refitting. Fifty null datasets are independently refitted and evaluated on the unchanged held-out graph groups. Confidence intervals use 1,000 relation-graph bootstrap replicates. The earliest real-history cutoff above the null R^2 95th percentile is Qwen layer 24 ($R^2 = 0.574$, AUROC $= 0.857$), Llama layer 14 ($R^2 = 0.393$, AUROC $= 0.837$) and Gemma layer 21 ($R^2 = 0.714$, AUROC $= 0.984$). This is a bounded prospective forecast of a declared candidate contrast. It is not an autonomous state equation or a general model of answer correctness.

9. Direct hidden-state chord geometry

For normalized hidden states, the endpoint chord is $g = x_L - x_1$. Chord alignment is the mean cosine between each step s_l and g. Path length is the sum of adjacent cosine distances, endpoint distance is the first-to-last cosine distance, detour is their ratio, and turning curvature is the mean angle between adjacent steps.

Each null retains x_1, x_L and the complete intermediate state set but randomly permutes intermediate layer order. One permutation is shared across all 96 prompts within a checkpoint, and 50 permutations are evaluated. The 50 checkpoint-level means form the null distribution; prompt-level p values are not pooled. Alignment gain is real minus shuffle mean. Detour and curvature reductions are $\log_2(\bar{q}_{\text{shuffle}}/\bar{q}_{\text{real}})$. Positive reductions favour observed order.

For a metric where larger values favour observed order, the null-standardized effect is

$$Z_q = \frac{q_{\text{real}} - \text{mean}(q_{\text{null}})}{\text{sd}(q_{\text{null}})}.$$

For detour and curvature the numerator is reversed so that positive values again favour observed order. The plus-one empirical one-sided p value is $(1 + n_{\text{extreme}})/(50 + 1)$. All three observed checkpoint means were beyond all 50 corresponding alignment null means, giving the minimum attainable empirical value of $1/51$; this finite resolution is reported rather than interpreted as a more precise p value. The standardized separation was 5.11, 5.57 and 5.63 null standard deviations for alignment; 11.87, 6.31 and 11.50 for detour; and 6.03, 4.20 and 6.42 for curvature in Qwen, Llama and Gemma, respectively.

Algorithm 3: endpoint-preserving layer-order null

```
for each checkpoint:
    compute one real checkpoint mean from the 96-prompt controlled subset
    repeat 50 times:
```

```
        draw one permutation of intermediate layer indices

        apply that same permutation to every prompt

        retain the first and final layer states

        compute the checkpoint mean for each geometry metric

    compare the real mean with the 50 repeat-level means
```

The estimand is limited to normalized hidden states, cosine geometry and this endpoint-preserving null. It does not test an exact geodesic equation. A fixed-seed rerun reproduced the archived output hashes; the hashes and filenames are recorded in the Source Data manifest.

10. Independent task-family and random-initialization controls

10.1 Controlled lexical and addition tasks

The independent task family asks which of two natural words belongs to a declared semantic category. It contains 64 problem groups and seven category inventories: colours, directions, metals, shapes, landscape features, sky objects and fruits. Candidate words were drawn from a 28-word intersection that tokenized as one continuation token in Qwen, Llama and Gemma. Correct-candidate display position alternated by problem identifier. Two stable prompts, two prompts containing a competing annotation and two correction prompts were generated for each problem, giving 384 prompts per checkpoint. The prompts used each checkpoint's native chat template.

The external task inherited the frozen relation-graph decision windows: layers 20-25 in Qwen, 12-15 in Llama and 18-23 in Gemma. No layer scan or task-specific window selection was performed. Problem groups were divided once with seed 20260714 into 44 training and 20 held-out groups. For each problem, the clean prompt supplied the candidate-margin baseline. PC1 was fitted to the training-group clean-relative decision profiles and applied to the held-out groups. A balanced multinomial logistic classifier used only this one-dimensional coordinate. Its held-out macro F1 was compared with 200 fits in which training labels were permuted while held-out labels and the split were unchanged.

The direct hidden-state geometry analysis retained all 384 prompts and used 50 endpoint-preserving intermediate-layer permutations per checkpoint. Real-order alignment exceeded all 50 null means in each checkpoint. Held-out lexical-coordinate macro F1 exceeded the label-null 95th percentile in Qwen and Llama but not Gemma. Clean-condition candidate selection was 0.891, 0.547 and 0.953, respectively, so the Llama result is retained with a weak-behaviour boundary.

The controlled addition task contains 64 single-step addition problems and the same six-condition design. It uses the same frozen decision windows, 44/20 problem split, train-only PC1 construction, 200 label shuffles and 50 endpoint-preserving order shuffles. Geometry exceeded all order nulls in all checkpoints. Held-out coordinate macro F1 was 0.583, 0.456 and 0.570 in Qwen, Llama and Gemma, above the corresponding label-null 95th percentiles. Llama pair accuracy was 0.315, so the coordinate result does not imply strong arithmetic behaviour. An earlier mixed-operation version is retained as a coverage audit rather than a primary result.

10.2 Standard four-choice ARC control

The standard benchmark control used the allenai/ai2_arc ARC-Challenge validation split. The 96-item audit subset comprised the earliest validation rows in dataset order whose answer keys were exactly A-D. This deterministic eligibility rule was fixed without reference to checkpoint outputs, bounding the extraction budget while preventing outcome-based item selection. Every original question and all four answer choices were presented through the checkpoint-native chat template, and the model was instructed to return one option letter. No binary clean/conflict wrapper was introduced. Four-choice accuracy was 0.781, 0.333 and 0.771 in Qwen, Llama and Gemma.

Hidden states were extracted using the same last-token rule as the main analysis. For each checkpoint, 50 endpoint-preserving intermediate-layer shuffles supplied the null. Real-order alignment exceeded all 50 null means, with standardized separations of 3.21, 5.23 and 7.84. The deterministic subset is an audit set rather than a new benchmark estimate. This analysis tests the order geometry beyond binary prompt structure; it does not transfer ΔU, mechanism labels or the intervention policy.

10.3 Random-initialization control

The random control used the exact Qwen2.5-1.5B, Llama-3.2-1B and Gemma-2-2B model configurations and native tokenizers. Three models per configuration were instantiated with seeds 1701, 1702 and 1703; token embeddings, attention and MLP parameters, normalization parameters and output heads were random, and no trained tensor was loaded. The inputs were the same fixed 96 raw relation-graph prompts used in the trained geometry audit. Hidden states were extracted after every transformer block using the same last-token rule.

Each parameter state was compared with 50 endpoint-preserving layer-order permutations using the same metric definitions and null seed 20260717. Mean random alignment gains were 0.0768, 0.1028 and 0.0879 in Qwen, Llama and Gemma, versus 0.0317, 0.0403 and 0.0499 in the trained checkpoints. Fig. 4 used a separately seeded 50-shuffle null; its small gain differences arise from the sampled null mean, while the trained real-order alignment is identical. All nine random gains were positive and exceeded their matched trained checkpoint. The result establishes an architecture-compatible order effect across these configurations. It does not show that trained and random trajectories carry the same function or identify a universal architecture-training decomposition.

11. Auxiliary readback-centre geometry

This auxiliary analysis operates on normalized TopK/VIM centres C_l, not raw hidden states. It uses the same broad family of endpoint-chord, cosine-path, detour and turning diagnostics and fixes endpoints while shuffling intermediate readback-centre layers. It is retained as a cross-observable diagnostic, and its estimand is never labelled hidden-state geometry.

12. Intervention labels, features and splits

Intervention-class labels are fixed by prompt condition in the separately generated ten-condition control panel. Stable contains clean, redundant, irrelevant and paraphrase; competition contains weak-distractor, balanced-competition and direct-conflict; closure contains closure-update, closure-override and exception-override. These are controlled intervention families rather than correctness labels or trajectory-derived dynamical phases. Their mapping is not substituted into the 12-condition coordinate audit.

Mechanism features are the predicted candidate-path separation and its magnitude, a companion dominance estimate, velocity magnitude and standardized velocity, normalized layer depth, a precursor-layer indicator, and baseline separation and dominance values. Policy features add mechanism probabilities and predicted-mechanism indicators. Training and test partitions are separated by relation graph, preventing rows from one graph from appearing in both partitions. Final correctness is not a target.

The first two intervention families used additive state directions, and the third used an additive flow direction. Their failure to yield robust mechanism-specific selectivity motivated the switch to a local mechanism-conditioned transition operator. At each intervention layer, principal-component analysis was fitted to the concatenated current and next hidden states from training graphs. Its dimension was $r_l = \min(128, 2n_{\text{train}} - 1, d_m)$. In the common PCA coordinates, separate stable and closure maps solved

$$(A_l^{(k)}, b_l^{(k)}) = \underset{A,b}{\operatorname{argmin}} \sum_{i \in \mathcal{T}_k} \|z_{i,l+1} - Az_{i,l} - b\|_2^2 + 10\|A\|_F^2,$$

for $k \in \{\text{stable}, \text{closure}\}$. The intercept was not penalized. The stable-row prediction was inverse-transformed to hidden-state space to define $\hat{F}_l^{\text{stable}}$. For realized next-layer state $h_{p,l+1}$, the intervention is

$$\tilde{h}_{p,l+1} = (1-\alpha_{p,l})h_{p,l+1} + \alpha_{p,l}\hat{F}_l^{\text{stable}}(h_{p,l}),$$

$$h_{p,l+1}^{\text{int}} = \tilde{h}_{p,l+1} + \beta_{p,l}\,\text{sd}(\tilde{h}_{p,l+1})d_l,$$

where d_l is the unit training-graph ridge direction for ΔU. The operator interpolation is applied first. The action grid is

$$\mathcal{A}_0 = \{(0,0),(0.1,0.1),(0.2,0.2),(0.3,0.3),(0.3,0),(0,0.3),(0.3,0.2),(0.2,0.3)\},$$

and `none` denotes $(0,0)$. For training prompts, closure labels maximize signed ΔU shift; stable and competition labels minimize absolute shift. The final policy is compared with matched fixed combinations and 50 mechanism-label, policy-label and joint shuffles.

The mechanism classifier used the nine-dimensional vector

$$x^{\text{mech}} = (\widehat{\Delta U}, |\widehat{\Delta U}|, \widehat{D}_B, \|v\|, z_v, \tilde{l}, I_{\text{precursor}}, \Delta U_{\text{base}}, D_{B,\text{base}}).$$

It was a 250-tree `RandomForestClassifier` with $max_depth = 7$, $min_samples_leaf = 6$, $class_weight = balanced_subsample$ and seed $42 + s$, where $s = 0$ for the real fit and the shuffle index otherwise. The policy input appended the three predicted mechanism probabilities and three one-hot predicted-mechanism indicators, giving $x^{\text{policy}} \in \mathbb{R}^{15}$. The policy was a 200-tree random forest with $max_depth = 6$, $min_samples_leaf = 8$, the same class weighting and seed $142 + s$. Both used the scikit-learn default weighted-Gini split criterion. They contained no neural hidden layers and used neither gradient descent nor policy gradients.

For every training prompt, each action $a = (\alpha, \beta)$ in the eight-element grid was executed. The discrete target was

$$a_p^* = \begin{cases} \text{argmax}_a \, [\Delta U_p^{(a)} - \Delta U_{p,\text{base}}], & p \in S_{\text{closure}}, \\ \text{argmin}_a \, |\Delta U_p^{(a)} - \Delta U_{p,\text{base}}|, & p \notin S_{\text{closure}}. \end{cases}$$

The policy forest learned this discrete label on layer-expanded training rows and predicted one action per held-out prompt-layer row. Thus the policy is supervised action classification after an exhaustive training-grid evaluation, not reinforcement learning.

The mechanism-label null permutes training mechanism labels, refits the mechanism classifier and then refits the downstream policy. The policy-label null retains the real mechanism classifier but permutes training action labels before policy fitting. The joint null independently permutes both label sets and refits both classifiers. All nulls retain the graph split, test rows, transition operators, action grid and endpoint calculation.

Executable procedure: held-out mechanism-conditioned intervention

```
split complete relation graphs 70:30 into training and held-out sets

fit transition operators, mechanism classifier and action policy on training graphs

for each held-out prompt and selected layer:

    predict mechanism probabilities

    select operator and precursor weights

    apply operator interpolation, then the scaled precursor update

    measure post-intervention Delta U and the declared candidate logits

compare the learned policy with matched fixed actions and refitted label-shuffle nulls
```

Specificity is

$$s(m,c) = \text{mean}_{i \in S_{\text{closure}}} |\Delta U_i^{(c)} - \Delta U_{i,\text{base}}| - \text{mean}_{i \in S_{\text{nonclosure}}} |\Delta U_i^{(c)} - \Delta U_{i,\text{base}}|.$$

The pseudo-log transform with $\sigma = 0.08$ is used only for Fig. 5 display. The primary endpoint of this original action-budget analysis is internal dominance specificity. Because the intervention is applied upstream of this held-out endpoint and is compared with fixed and refitted label-shuffle controls, it supports selective causal control of the measured internal answer-formation trajectory. Output-boundary control is tested separately below with an expanded, training-only action search and a strict full-vocabulary endpoint.

The candidate-choice boundary pairs no-intervention and real-policy outputs for each held-out prompt. It reports the change in final clean-minus-conflict candidate margin, the change in binary choice between those two candidates and their association with ΔU shift. Intervals for these paired descriptive contrasts use 5,000 relation-graph bootstrap replicates with seed 20260714. In closure prompts, clean denotes the original unperturbed-chain candidate, whereas the updated or override-supported conflict candidate is generally the rule-consistent answer. A conflict-to-clean crossing is therefore an actuator-response endpoint, not an accuracy improvement.

Out-of-fold margin-shift and boundary-crossing analysis

The quantitative bridge defines

$$\delta M_p = M_{p,\text{policy}} - M_{p,0}, \qquad \delta\Delta U_p = \Delta U_{p,\text{policy}} - \Delta U_{p,0}.$$

For checkpoint m and regime r, an affine ridge model with penalty 10^{-6} estimates

$$\widehat{\delta M}_p = a_{m,r} + b_{m,r}\delta\Delta U_p.$$

Five-fold `GroupKFold` holds out complete relation graphs, with the same graph held out across checkpoints. The predicted post-policy margin is $\widehat{M}_{p,\text{policy}} = M_{p,0} + \widehat{\delta M}_p$. For the at-risk population $M_{p,0} < 0$, the observed crossing label is $I(M_{p,\text{policy}} \geq 0)$. Within each outer fold, an inner graph cross-fit supplies training predictions for a one-variable logistic calibrator, which maps predicted post-margin to crossing probability. Thus neither the margin-shift prediction nor its probability calibration evaluates a graph used for fitting.

The pooled closure margin-shift model reached $R^2 = 0.850$ with a 95% graph-bootstrap interval of 0.811-0.882. Among 188 at-risk closure prompts, 15 crossed the candidate boundary and pooled AUROC was 0.976 (0.951-0.996). Under the original action budget, the observed counts were 0/87 in Qwen, 9/45 in Llama and 6/56 in Gemma. Qwen's mean shift was largest in absolute logit units, but its mean initial deficit was 5.03; 94.0% of its closure-layer actions used the maximum $(0.3, 0.3)$ action. Continuous and crossing intervals use 1,000 graph-bootstrap replicates with seed 20260714. Internal trajectory control was observed in all three checkpoints, whereas candidate-boundary crossing under the original action budget was checkpoint-dependent. This endpoint is not open-ended generated-answer correctness.

Confidence-gated emitted-token boundary control

The expanded output-boundary analysis was run independently in Qwen, Llama and Gemma with the same seed-42 split of complete relation graphs: 67 training graphs containing 670 prompts and 29 held-out graphs containing 290 prompts, of which 87 were closure prompts. Intervention layers were 15-19 in Qwen, 8-11 in Llama and 13-17 in Gemma. The principal-component direction defining ΔU, transition operators, precursor directions, mechanism classifier, action targets and boundary policy were fitted on training graphs only. Candidate token identities entered the paired-margin objective. Generated answer text, final-answer correctness and held-out candidate outcomes entered neither fitting nor gate construction.

The bounded action library was

$$\mathcal{A} = \{(0,0), (0.3, 0.3), (0.6, 0.3), (0.6, 0.6), (0.9, 0.3), (0.9, 0.6), (0.9, 0.9), (1.0, 0.6), (1.0, 0.9), (1.0, 1.2)\}.$$

For action $a = (\alpha, \beta)$, the state update retained the operator-then-precursor order defined above. Because $0 \leq \alpha \leq 1$, operator interpolation could reach but not extrapolate beyond the fitted transition proposal. On a training closure prompt p, action utility was

$$u_p(a) = \delta M_p(a) + 5I[M_{p,a} \geq 0] - 0.03(\alpha^2 + \beta^2),$$

where $\delta M_p(a)$ is the change in clean-minus-conflict candidate margin and the indicator supplies a crossing bonus. For non-closure prompts,

$$u_p(a) = -|\delta M_p(a)| - 0.10(\alpha^2 + \beta^2).$$

The policy was trained to predict the utility-maximizing action from the same 15 pre-intervention features used above. This objective searches for a bounded action that crosses the declared candidate boundary in closure while penalizing movement and dose outside closure; it does not optimize correctness.

For prompt p, let $\pi_{p,l}(k)$ be the intervention-class posterior for class k at intervention layer $l \in L_I$. Mean closure support and normalized posterior entropy were

$$\bar{\pi}_p = |L_I|^{-1} \sum_{l \in L_I} \pi_{p,l}(\text{closure}), \qquad \bar{H}_p = |L_I|^{-1} \sum_{l \in L_I} \frac{-\sum_k \pi_{p,l}(k) \log \pi_{p,l}(k)}{\log 3}.$$

The gate was

$$g_p = I[\bar{\pi}_p \geq 0.50 \ \wedge \ \bar{H}_p \leq 0.90].$$

The probability and entropy thresholds were declared in the experiment protocol before held-out execution. No grid search, threshold sweep or held-out outcome tuning was performed. The reported result therefore characterizes the operating point $(0.50, 0.90)$, not an optimized gate or a threshold-robustness claim. When $g_p = 0$, every intervention-layer action was set to $(0, 0)$. When $g_p = 1$, the predicted model-specific action vector was executed. The primary endpoint was a strict full-vocabulary top-1 continuation change from the declared conflict token to the declared clean token. The damage guards were any full-vocabulary top-1 change among the 203 non-closure prompts and intervention-layer state-norm ratios. Clean denotes the unperturbed-chain candidate and is generally not the rule-consistent answer in closure prompts.

The confidence-gated controller crossed the declared boundary in 17/87, 13/87 and 9/87 held-out closure prompts in Qwen, Llama and Gemma. Non-closure top-1 changed in 0/203, 2/203 and 0/203 prompts, respectively. Without the final abstention gate, strict crossings were 19/87, 13/87 and 9/87, while non-closure changes were 3.45%, 1.48% and 0.49%. These results support output-boundary causal leverage at one declared operating point across the three testbeds. They do not establish answer-correctness improvement, a safety guarantee or deployment control.

To test whether the effect required precise action-to-trajectory assignment, each safety-matched repeat permuted the complete model-specific action vector without fixed points among gate-open prompts within the same closure condition. This preserved the gate-open set, all gate-closed zero actions, total action budget and every condition-by-layer (α, β) multiset. Strict crossings were defined relative to each prompt's unperturbed top-1 output. Across 50 repeats, null crossing counts ranged from 15 to 20 in Qwen, 12 to 16 in Llama and 6 to 10 in Gemma. Their means were 17.64, 14.18 and 7.90. The one-sided empirical values were 0.804, 0.922 and 0.255, respectively.

The matched null therefore does not support fine-grained action-to-trajectory dose specificity in any checkpoint. A separately frozen component-ablation protocol retained the same gate-open sets and training fits. Operator-only actions kept each real gated α and set $\beta = 0$; precursor-only actions kept β and set $\alpha = 0$; the uniform control assigned $(1.0, 1.2)$, the maximum existing library action, to every gate-open prompt. Negative and positive references reproduced all previous top-1 token IDs exactly before the ablations were interpreted.

Supplementary Methods Table 3 | Confidence-gated actuator component attribution.

Checkpoint	Gated policy: closure flips / non-closure changes	Operator only	Precursor only	Uniform maximum on same gate
Qwen	17/87 / 0/203	17/87 / 0/203	0/87 / 0/203	22/87 / 0/203
Llama	13/87 / 2/203	14/87 / 2/203	1/87 / 0/203	15/87 / 2/203
Gemma	9/87 / 0/203	8/87 / 0/203	0/87 / 0/203	10/87 / 1/203

Across checkpoints, the gated policy and operator-only ablation each produced 39 strict crossings and 2 non-closure changes; precursor only produced 1 and 0, and the uniform maximum produced 47 and 3. All intervened states were finite and the largest recorded state-norm ratio was below 1.10. The supported algorithmic object is therefore more specific than the original two-component description: confidence-based abstention plus the bounded local transition operator carries the principal output-boundary leverage at the declared operating point. The precursor component alone is insufficient. The Llama and uniform-Gemma damage counts show that the same gate does not guarantee zero collateral output change across checkpoints. These comparisons do not identify a universal decomposition or establish correctness control.

Executable procedure: confidence-gated local transition control

```
split complete relation graphs into training and held-out sets

fit the candidate-separation coordinate, transition operators and classifiers on training graphs

execute every bounded action on training prompts and assign the utility-maximizing action label

for each held-out prompt:

    compute layerwise mechanism posteriors from pre-intervention features

    open the gate only if mean closure support is at least 0.50 and mean entropy is at most 0.90

    if the gate is closed, apply the identity action at every intervention layer

    if the gate is open, apply the predicted bounded operator and precursor weights

    evaluate strict full-vocabulary top-1 candidate crossing and non-closure output change

compare the real assignment with safety-matched action-vector reassignments
```

Frozen mixed-arithmetic actuator-transfer test

This experiment asked whether the already specified actuator procedure transferred to a second task after training-only refitting. It was not a zero-shot transfer of numerical operators or of the relation-graph ΔU axis. The empirical object remained the ordered last-token layer-state process. The observable endpoint was the distractor-versus-arithmetic-consistent candidate margin and the full-vocabulary top-1 token before and after intervention.

The task contained 64 unique expressions sampled deterministically across addition, subtraction and multiplication. Candidate answers were number words from zero to twenty, counterbalanced by problem parity. Every problem supplied six conditions: clean and paraphrase mapped to stable; untrusted conflict and asserted conflict mapped to competition; explicit correction and final recheck mapped to the target correction class. A single group split with seed 20260714 assigned 44 complete problems to training and 20 to held-out evaluation. No held-out row selected a layer, action, threshold, regularization value or stopping rule.

The primary assay used a raw prompt ending immediately before the requested one-word continuation, matching the relation-graph actuator interface. For every checkpoint, tokenized prompt-plus-candidate text had to contain the complete prompt token sequence as an exact prefix, and the first appended token had to equal the audited single leading-space token for that candidate. All labels from zero to twenty passed this gate. An initial non-contextual token run failed this check and is excluded. A subsequent native-chat pilot passed contextual tokenization but is not pooled with the primary result: Llama began all 120 held-out continuations with free text rather than either candidate, so a candidate-token top-1 crossing was outside that interface's endpoint domain. These pilots diagnose interface validity and are not intervention evidence.

The following quantities were inherited without retuning from the relation-graph boundary experiment: Qwen, Llama and Gemma intervention and decision windows; the local transition-operator parameterization and ridge penalty; the ten-action (α, β) library; mechanism and action-classifier hyperparameters; closure-probability threshold 0.50; normalized-entropy threshold 0.90; and the 5% non-target collateral ceiling. Task-specific arithmetic coordinates, numerical transition maps, precursor directions and action labels were fitted only on the 44 training problems. Thus the frozen object was the intervention procedure and hyperparameter contract, whereas fitted numerical coordinates were allowed to adapt to the new task.

The primary control was the confidence-gated learned policy. Negative and attribution controls were no intervention, ungated policy, training-selected fixed global action, operator only, precursor only and uniform maximum action. For each checkpoint, 50 address-destruction nulls randomly reassigned complete action bundles across held-out prompts. Every repeat preserved the number of open gates and the exact multiset of layerwise (α, β) doses; 150 of 150 model-repeat dose-multiset checks passed. This null destroys prompt-to-action address while retaining actuator availability and aggregate budget. It does not test fine dose assignment among already addressed targets.

Strict eligibility required a held-out correction prompt whose unperturbed full-vocabulary top-1 token was the distractor candidate. A strict crossing required the intervened full-vocabulary top-1 to become the arithmetic-consistent candidate. Non-target damage was any top-1 change among stable and competition prompts. For checkpoint m,

$$s_m = \frac{N_{m,\text{cross}}}{N_{m,\text{eligible}}} - \frac{N_{m,\text{non}-\text{target change}}}{N_{m,\text{non}-\text{target}}}.$$

If $N_{m,\text{eligible}} < 5$, the checkpoint was declared underpowered for a model-specific crossing verdict. Confirmation required: (i) at least two checkpoints with at least five eligible prompts, positive median eligible candidate-margin shift and one or more strict crossings; (ii) at least two checkpoints with collateral at or below 5%; (iii) pooled specificity above the 95th percentile of 50 pooled address-null repeats; and (iv) a non-zero pooled crossing count with no numerical-instability failure. Partial internal or pair-boundary transfer was the predeclared revised verdict when only part of this gate passed.

The primary results are in Supplementary Methods Table 6. Qwen was the only checkpoint to combine a powered denominator, positive median margin shift and a strict crossing. Llama crossed once but had a negative median shift; Gemma had a positive median shift but only four eligible prompts and no primary crossing. Pooled strict crossings were 2/45, non-target changes 1/240 and specificity 0.04028. The pooled null 95th percentile was -0.01111 and the plus-one one-sided empirical value was $1/51 = 0.0196$. The address result therefore supports selective routing at the pooled level, but the two-checkpoint positive-direction gate failed. The frozen decision is partial protocol-level transfer, not confirmed cross-task output control.

The arithmetic PC1 coordinate did not provide a universal intervention axis. In Qwen, the median eligible candidate-margin shift was positive (0.5391) while the median task-coordinate shift was negative (-0.0206); Llama was negative on both, and underpowered Gemma was positive on both. These mixed directions are retained because strict output evidence must be evaluated at the declared token boundary rather than inferred from a compressed coordinate. Operator-only reproduced the two primary crossings, precursor-only produced none and uniform maximum produced two crossings plus one additional Gemma crossing with one Gemma non-target change (Supplementary Methods Table 7). These controls do not establish fine action assignment or a task-general executor.

For execution reproducibility, full-data extraction and held-out replay were allowed a maximum absolute FP16 candidate-margin discrepancy of 0.125, with any sign disagreement confined to that near-zero band. An independent repeat using the identical held-out batch partition had to reproduce every full-vocabulary top-1 token and candidate-pair sign exactly. All three checkpoints passed. An independent verifier recomputed the primary summaries, pooled null, decision gate and all action-dose multisets from retained row-level files without importing the execution code.

13. Human-AI hypothesis-to-audit workflow

This section records research provenance and responsibility. The workflow was not evaluated as a productivity intervention: the development interval records achieved throughput in this case but does not identify the counterfactual contribution of AI assistance. The collaboration used a repeated six-step loop.

1. **Human question or contradiction:** identify a mechanistic question, evidentiary gap or failed interpretation.
2. **Executable proposal:** use AI assistance to translate the question into candidate metrics, controls, scripts or consistency checks.
3. **Local execution:** run analyses against stated checkpoints and archived data.
4. **Evidence audit:** compare outputs with leakage restrictions, nulls, source files and rival explanations.
5. **Human decision:** retain, revise or reject the object, metric, interpretation or claim.
6. **Trace-preserving iteration:** archive scripts, outputs, negative results and the reason for the next test.

AI tools assisted with project-record retrieval, code scaffolding and debugging, batch planning, data-table organization, deterministic R plotting scripts, consistency checks and language editing. The human researcher selected the scientific problem, changed the empirical object, decided which controls were decisive, interpreted failures, approved claim boundaries and remained accountable for every reported output. AI suggestions were operational drafts or candidate hypotheses, not evidence.

The primary trajectory-mapping, cross-checkpoint, geometric and intervention code sequence was developed from 2 to 6 June 2026. A proxy audit on 10 June separated direct hidden-state, vocabulary-facing and construction-aligned objects and forced correction of earlier interpretations. These dates document development order and the achieved interval, but they do not quantify productivity relative to a no-AI process, establish scientific quality or mark completion of every final run. Final archive metadata should additionally preserve execution times, hashes and environments.

14. Responsibility matrix

Function	Human authority	AI-assisted operation	Auditable artefact
Problem and object	Define question; select and revise process object	Retrieve prior discussions; compare formulations	Conversation archive and provenance ledger
Theory	Select source ideas and limits; reject misleading analogies	Organize concept mappings and draft formulas	Concept-provenance table
Experiment	Specify controls, nulls and acceptance criteria	Draft code scaffolds and parameter sweeps	Versioned scripts and configuration tables
Falsification	Interpret contradictions and decide theory change	Locate inconsistencies and propose follow-up checks	Negative-result and correction records
Data and figures	Select quantities; verify source mapping; approve claims	Assemble tables and deterministic R scripts	Source Data manifest and figure audit
Manuscript	Set evidence hierarchy, scope and final language	Terminology, formatting and language support	Versioned manuscript and audit report
Accountability	Approve and assume responsibility	No approval or authorship authority	Author Contributions and AI disclosure

15. Research independence and disclosure completeness

The study was undertaken independently and was not an internal project of any listed organization. It used no employer data, computing infrastructure, internal systems, funding, research facilities or personnel support and did not involve organizational review or approval. H.Z. personally provided or paid for all local computing, storage, software subscriptions and other research expenses. Y.W., R.S. and W.C. contributed in their personal capacities as independent research collaborators. Affiliations identify the authors and do not imply employer sponsorship, endorsement or responsibility for the work.

The author names, affiliations, corresponding-author details, ORCID, acknowledgements, funding statement, narrative author contributions and competing-interests declaration were carried forward from the preprint record and checked against the main manuscript. The submitted scientific figures were generated from retained data and deterministic R code rather than an image-generation system.

AI product families and providers are disclosed in the main Methods. The consumer interfaces used during June and July 2026 did not consistently expose stable underlying model build identifiers, and interface-level task identifiers were not retained as scientific provenance; these fields are therefore not reconstructed. Scientific provenance instead rests on the executable scripts, fixed inputs, environment records, null specifications, retained outputs, source mappings and file-level hashes included in the archive. The public repository and Zenodo concept records are identified in the main Data and Code availability statements. Formal CRediT labels are supplied as submission metadata and do not replace the narrative author-contribution statement.

16. Additional architecture, order, metric and training controls

16.1 Random fixed-map relational continuity

Prompt states were propagated through fixed random mappings matched to the dimensional schedules of representative Qwen, Llama and Gemma configurations. The audit crossed three task families (lexical categories, addition and four-choice ARC-Challenge) with two independent seeds. The tested quantity was a cross-depth relation-preservation contrast computed from independently partitioned prompt sets and compared with lag-matched, block-donor and independent-map nulls. All 18 declared architecture-task-seed conditions passed the positive-contrast gate. The accepted claim is architecture-compatible relational continuity. Learned semantics, native-order privilege, endpoint direction and a least-action path law do not follow from this result.

16.2 Reordered execution and functional fidelity

The same trained Qwen checkpoint was executed in native, reversed and fixed-permutation block order. Geometry was compared with native full-vocabulary top-1 output, declared-candidate agreement and Jensen-Shannon divergence. Reversed and fixed-permutation executions retained positive chord contrasts but had zero full-vocabulary top-1 agreement with native execution; candidate agreement was 0.479 and mean Jensen-Shannon divergence was approximately 0.692. This falsified geometric straightness as a sufficient statistic for native computation. The retained claim is that realized order is functionally consequential even when a coarse geometric statistic remains positive.

16.3 Incremental path-history prediction

For each task-checkpoint condition, group-held-out current-state predictors were compared with otherwise matched predictors augmented by ordered-prefix summaries. Passing required a positive incremental held-out gain above the declared order-null threshold. Six of 18 initial conditions passed, all in addition; lexical prompts passed 0/9. An independently generated mixed-arithmetic replication passed 0/9. The relation-graph final-margin forecast remains valid within its declared construction, but the tested prefix summaries do not establish a task-general path-memory law. A compatible explanation is that the current high-dimensional state already compresses much of the usable history; this explanation has not yet been tested as an exact Markov property.

16.4 Radial sensitivity of the Llama statistic

Llama residual trajectories were decomposed into unit directions and state norms. Direction-only trajectories passed none of seven task gates, whereas norm-only fixed-direction trajectories passed all seven. A frozen radial exponent $\gamma \in \{0, 0.25, 0.5, 0.75, 1, 1.25\}$ produced monotonic metric changes in all seven tasks. The accepted claim is radial sensitivity under the declared metric and normalization. The statistic does not independently identify semantic direction, function or an architecture-wide law.

16.5 Pythia component compatibility

Embeddings (E), transformer blocks (B), final normalization (N) and output head (W) were swapped between initialized and trained Pythia-160m checkpoints in the complete 2^4 factorial design. All combinations were evaluated on the same 53 held-out lexical cloze prompts. Accuracy rose from 0.472 with all initialized components to 0.849 with all trained components; the best isolated trained component reached 0.604. The fully trained margin exceeded the additive component prediction by 2.621 (95% bootstrap confidence interval, 1.252-3.987). The supported claim is distributed compatibility among trained input, transition and output components in this checkpoint and task. Hybrid combinations are diagnostic interventions, not natural training states.

16.6 Training-time task coordinates

Seven Pythia checkpoints were evaluated with a model-native candidate coordinate and a coordinate fitted once at the final checkpoint. Model-native order gain was absent at initialization, appeared from step 1,000 and correlated with held-out accuracy (Spearman $\rho = 0.75$). The fixed final coordinate transferred with macro F1 0.041 at steps 0 and 128, 0.701 at step 1,000 and 1.0 by step 16,000. A coordinate refitted within each checkpoint already separated categories perfectly at initialization and therefore failed the learning-evidence gate. These results support training-associated task organization in fixed coordinates while rejecting random-initialization linear separability as evidence of learned semantics.

Supplementary Methods Table 4 | Claim-boundary audit.

Supported statement	Boundary
Ordered hidden states provide a common within-checkpoint process object	Engineering components still implement the process; cross-checkpoint coordinates are not assumed identical
Direct hidden and hidden-flow features provide stronger mechanism coordinates than readbacks in the tested audits	Readbacks remain useful diagnostics; the construction-aligned decision profile is not independent evidence
Exploratory functional windows retain prompt-condition-class information on new graph instantiations	The joint prospective recurrence gate failed: topology separation reversed in all checkpoints, downstream coupling missed its null in Llama and only Gemma retained top-quartile window location
ΔU summarizes candidate-path separation	It is construction-defined, not an energy, final-answer label or universal order parameter; in the frozen predecision audit it did not exceed matched recent logit margins in any checkpoint
O_l^{cont} improves closure	It is descriptive when adjacent target-layer information enters its construction; corrected target-excluded bilinear estimates retained 25.1% of the observed-coordinate gain and remain an exploratory predictive boundary
Hidden paths are chord-directed relative to endpoint-preserving shuffles	The result is metric-dependent and does not establish exact geodesics or least action; the new-graph Gemma-3-12B result is a larger independent checkpoint confirmation, not a pure scale law
Chord-directed geometry recurs in lexical, addition and four-choice ARC tasks	Task-specific coordinate transfer is partial, and the numerical relation-graph ΔU axis was not transferred
Ordered partial histories forecast the final candidate margin in held-out relation-graph groups	Cross-task ordered-prefix summaries did not generally add prediction beyond current state; no universal path-memory law is supported
Layer order yields chord-directed geometry in trained and randomly initialized Qwen, Llama and Gemma architectures	Random-initialization effects were larger; geometry alone is architecture-compatible and does not identify learned inference organization
Fixed random maps preserve prompt relations across depth under three stronger nulls	Relation preservation does not establish semantics, native-order privilege or task function
Reverse and permuted executions retain positive geometry but lose native outputs	Native order is functionally consequential; straightness is not sufficient for fidelity
Pythia task function depends on trained $E/B/W$ compatibility and fixed task coordinates emerge during training	One small model, one training run and one lexical diagnostic do not establish a universal training law or complete representational geometry
Llama raw alignment is strongly radial-sensitive	Metric sensitivity does not make norm scale the mechanism or establish semantic direction
Local transition operators shift internal dominance beyond fixed and shuffled controls	Internal selectivity does not by itself establish an emitted-token change
Held-out internal displacement predicts candidate-margin movement and pooled crossing risk	Original-budget crossings were checkpoint-dependent; the map is task- and candidate-defined
Confidence-gated local transition control crosses the emitted-token candidate boundary in all three checkpoints	Strict flips were 17/87, 13/87 and 9/87; operator-only ablation preserved 17/87, 14/87 and 8/87 whereas precursor-only produced 0/87, 1/87 and 0/87. A matched empirical-subspace direction null did not support direction specificity after correction across the three checkpoints; this is not answer correctness, safety or deployment control
The frozen actuator procedure shows pooled selectivity in mixed arithmetic	Only Qwen passed the model-level positive-direction gate; Llama's median target shift was negative and Gemma was underpowered. The two-model confirmation gate failed, so this is partial protocol transfer rather than cross-task output control
An AI-assisted workflow completed a compressed hypothesis-to-audit sequence	This single case has no no-AI counterfactual and does not identify a causal productivity gain; scientific object choice, falsification decisions, interpretation and accountability remained human

Supplementary Methods Table 5 | Held-out closure-prompt boundary outcomes under confidence-gated control.

Strict-crossing eligibility required the baseline full-vocabulary top-1 token to be the conflict candidate. A pair-margin crossing means that the clean-minus-conflict margin became non-negative while another vocabulary token remained top-1. Categories are mutually exclusive and sum to 87 closure prompts per checkpoint.

Checkpoint	Eligibility	Mutually exclusive gate-open eligible outcomes	Abstention or ineligibility
Qwen	87/87 baseline-conflict; 84 gate-open	17 strict crossing; 67 positive but below pair boundary; 0 pair crossing behind third token; 0 non-positive shift	3 eligible gate-closed; 0 not baseline-conflict
Llama	44/87 baseline-conflict; 40 gate-open	13 strict crossing; 19 positive but below pair boundary; 6 pair crossings behind third token; 2 non-positive shifts	4 eligible gate-closed; 43 not baseline-conflict
Gemma	17/87 baseline-conflict; 17 gate-open	9 strict crossing; 5 positive but below pair boundary; 3 pair crossings behind third token; 0 non-positive shifts	0 eligible gate-closed; 70 not baseline-conflict

For Qwen, positive-shift non-crossings began farther from the pair boundary than strict crossings (median baseline margin, -5.58 versus -3.67), although their median shifts were similar (3.98 versus 4.41). This pattern is consistent with insufficient bounded displacement at the tested operating point rather than a general failure to move the candidate margin. In Llama and Gemma, six and three eligible prompts crossed the pairwise candidate margin but remained behind a third full-vocabulary token. The decomposition therefore separates gate abstention, insufficient pairwise displacement and third-token competition; it does not establish that increasing the action budget would preserve collateral-output rates.

Supplementary Methods Table 6 | Frozen mixed-arithmetic primary actuator result.

Eligibility requires the unperturbed full-vocabulary top-1 token to be the distractor candidate on a held-out correction prompt. Specificity is target crossing rate minus non-target top-1-change rate. The arithmetic coordinate was fitted on training problems and is not the relation-graph ΔU axis.

Checkpoint	Eligible prompts	Strict crossings	Median candidate-margin shift	Median arithmetic-coordinate shift	Non-target changes	Specificity	Model-level gate
Qwen	14	1	0.5391	-0.0206	1/80	0.0589	Pass
Llama	27	1	-0.0703	-0.0414	0/80	0.0370	Fail: median direction
Gemma	4	0	0.0938	0.5538	0/80	0.0000	Underpowered
Pooled	45	2	Not pooled	Not pooled	1/240	0.0403	Fail: only one model passes

Supplementary Methods Table 7 | Frozen mixed-arithmetic actuator-component controls.

Values are strict crossings among the same model-specific eligible prompts followed by non-target top-1 changes among 80 stable or competition prompts. The maximum action is the largest action already present in the frozen library.

Checkpoint	Primary guarded policy	Operator only	Precursor only	Uniform maximum
Qwen	1/14; 1/80	1/14; 1/80	0/14; 0/80	0/14; 1/80
Llama	1/27; 0/80	1/27; 0/80	0/27; 0/80	1/27; 0/80
Gemma	0/4; 0/80	0/4; 0/80	0/4; 0/80	1/4; 1/80

Supplementary Methods Table 8 | Experiment subset, window and null-control metadata.

Analysis	Subset / grouping	Window or ordering	Split / null	Primary estimand
Coordinate audit	960 traces per checkpoint; graph groups	Precursor and decision windows mapped by depth	Real, row-permuted and Gaussian projections	Mechanism-coordinate score and preservation
Functional registration	Full functional-window scan; graph groups	First 45% depth; widths 2, 3, 4, 5, 6 and 8; step 1	Group-aware folds inside non-nested exploratory selection	Preservation, prompt-condition-class and downstream objective
Prospective functional-window boundary	24 new graph groups; disjoint entities, relation phrases and candidate labels	Discovery-selected window and K frozen per checkpoint	Discovery-only fits; 2,000 graph bootstraps; 999 label and target permutations	Joint role recurrence and frozen-window rank
ΔU	Task-defined clean/conflict candidates	Decision window; predecision exclusion controls	Final correctness labels excluded	Oriented PC1 of clean-relative margin matrix
Direct hidden geometry	96 prompts: 8 complete groups x 12 conditions	Full observed layer order	50 fixed-endpoint intermediate-layer permutations	Alignment gain; log2 detour and curvature reduction
Mechanism-conditioned intervention	670 policy-train and 290 held-out prompts	Layer-expanded rows	Group-aware split by graph group; 50 mechanism, policy and joint shuffles	Internal dominance specificity
Transition predictive boundary	14,592 layer-transition rows; graph groups	Pre-transition state, history and context only	GroupKFold; target layer excluded	Retained fraction of observed-coordinate closure gain
Prospective final-margin forecast	670 training and 290 held-out prompts; graph groups	Current and earlier decision-window margins only	1,000 graph bootstraps; 50 independent within-prompt order nulls	Held-out final candidate-margin R^2 and boundary AUROC
Candidate-choice boundary	290 paired held-out prompts per checkpoint; 29 graph groups	No intervention versus mechanism-aware policy	5,000 relation-graph bootstrap replicates	Candidate-margin shift and clean-candidate selection change
Independent controlled tasks	Lexical and addition: 64 problems x 6 conditions per checkpoint	Main-study windows frozen before coordinate evaluation	44 train and 20 held-out problems; 200 training-label shuffles; 50 endpoint-preserving order shuffles	External chord alignment and held-out task-coordinate macro-F1
Standard four-choice task	ARC-Challenge validation: 96 items per checkpoint	Full four-choice prompts; no binary reduction	Deterministic subset; 50 endpoint-preserving order shuffles	Chord alignment and four-choice accuracy
Random-initialization geometry	Three seeds per Qwen, Llama and Gemma configuration; same 96 relation-graph prompts	Full observed layer order	50 endpoint-preserving intermediate-layer permutations per parameter state	Alignment gain relative to order-null mean
Held-out displacement-boundary bridge	290 intervention-test prompts per checkpoint; 29 graph groups; 188 at-risk closure prompts	Policy-induced ΔU and candidate-margin shifts	Out-of-fold graph-wise affine ridge fits; nested calibration; 1,000 graph bootstraps	Continuous margin-shift R^2, crossing AUROC and observed crossing counts
Confidence-gated output boundary	Each checkpoint: 670 training and 290 held-out prompts; 87 held-out closure prompts	Model-specific intervention windows; shared expanded bounded action library	Train-only coordinate and fits; 50 safety-matched action-vector reassignments per checkpoint	Full-vocabulary conflict-to-clean top-1 flips and non-closure output change
Frozen mixed-arithmetic actuator transfer	64 problems x 6 conditions; 44 train and 20 held-out problems per checkpoint	Relation-graph windows, action library, regularization and gate thresholds frozen; task-specific fits use training problems only	Raw-prompt token-boundary audit; 50 action-address reassignments per checkpoint; independent summary and dose-multiset verification	Strict distractor-to-arithmetic-consistent top-1 crossings, eligible margin shift, collateral change and predeclared cross-model verdict
Random-map relation continuity	2 new seeds x 3 architectures x 3 tasks; 96 prompts each	Cross-half and first-to-last-quarter depth	199 lag-1, contiguous-block and fingerprint-preserving donors	Unbiased linear CKA beyond all three nulls
Reordered execution	Trained Qwen; 48 prompts	Native, reverse and one frozen block permutation	10,000 paired bootstraps; 100,000 sign flips	Geometry-function dissociation
Incremental path memory	Lexical, addition and independent mixed arithmetic; 3 checkpoints x 3 horizons	Ordered prefix versus current state and unordered history	Five-fold problem groups; 199 order shuffles; multiplicity correction	Incremental balanced-accuracy gain
Llama radial audit	4 original and 3 independent tasks	Direction-only, norm-only and six frozen radial exponents	199 layer-conditioned radial donors	Scale sensitivity of raw alignment statistic
Pythia component compatibility	53 lexical prompts; step 0 and step 143,000	All 16 $E/B/N/W$ component combinations	5,000 paired bootstraps; full 2^4 factorial contrasts	Pair accuracy and superadditive margin excess
Training-coordinate audit	53 prompts x 7 Pythia checkpoints	Model-native candidate coordinate; held-out category coordinate	999 layer-order shuffles; 199 label nulls	Order gain, fixed-coordinate transfer and macro F1
Predecision coordinate competition	96 relation-graph groups per checkpoint	Frozen predecision cutoffs: Qwen 22, Llama 13 and Gemma 20	Five GroupKFold partitions; 100,000 graph-level sign flips; BH across 9 contrasts	OOF mechanism-label macro-F1 against matched recent logit margins
Empirical direction null	290 held-out relation-graph prompts per checkpoint	Published intervention layers and gate preserved exactly	50 empirical update-subspace random directions per checkpoint; BH across 3 checkpoints	Closure strict crossings minus non-closure full-vocabulary top-1 changes
New-graph geometry	24 new groups x 10 fixed conditions = 240 prompts per checkpoint	All native block states, no window selection	50 shared endpoint-preserving order shuffles; BH across 2 checkpoint alignment tests	Checkpoint mean chord alignment

17. Frozen closure audits and claim-family multiplicity

The closure package was specified before extraction to address three distinct potential alternatives: a cheap predecision readback could account for the proposed coordinate; bounded output crossings could arise from generic directions of matched magnitude; and geometry could be limited to the original graph instances and checkpoint scale. Its protocol fixed the prompt construction, cutoffs, group splits, endpoints, nulls and decision rules. It forbade subsequent window scans, policy searches and outcome-driven metric choices.

The three audits have different analysis units and null-generating mechanisms. Coordinate contrasts use 96 graph-level held-out contributions and 100,000 paired sign flips. Direction tests use a checkpoint-specific 290-prompt held-out intervention cohort and 50 matched empirical-subspace draws. Geometry uses one checkpoint-level mean from 240 prompts and 50 shared layer permutations. We therefore corrected only within the named evidence family and did not construct an uninterpretable study-wide FDR across geometry, grouped classification, paired output crossings and training interventions. The complete registry, including existing evidence families, is released as `code/extension_v0_56/multiplicity_claim_family_register_v0_56.csv`.

17.1 Predecision coordinate competition

At a frozen predecision cutoff, let R_l be the clean-minus-conflict output-logit margin and dR_l the same margin relative to the clean prompt in the graph group. The primary low-cost comparator was the matched recent vector $[dR_l]$ from the frozen two- or three-layer readback. The candidate coordinate ΔU_{pre} was a one-dimensional PC1 score derived only from the complete training-fold dR history through the cutoff. A classifier saw no post-cutoff state, final token, correctness label or held-out statistic. Scaling, PCA and class-balanced multinomial logistic regression were refitted in each GroupKFold training fold.

The planned gate for a distinct ordered-readback contribution required both ΔU_{pre} and the complete prefix to improve over the matched recent baseline after family correction. ΔU_{pre} was lower in all three checkpoints; the gate failed. The complete prefix and direct hidden state did improve, but the audit does not contain an unordered prefix control. The permitted conclusion is therefore that full predecision information can matter, not that layer order or ΔU adds an independent advantage.

Supplementary Methods Table 9 | Frozen predecision coordinate competition.

Values are out-of-fold macro-F1 differences relative to the matched recent clean-relative logit-margin vector. One-sided graph-level sign flips used 100,000 repetitions. BH correction is across all nine listed contrasts.

Checkpoint	ΔU_{pre}	Complete prefix	Cutoff hidden state	q for ΔU_{pre}	q for positive contrasts	Distinct-coordinate gate
Qwen	-0.0644	0.2166	0.3254	1.0000	1.50×10^{-5}	Fail
Llama	-0.1649	0.2996	0.5317	1.0000	1.50×10^{-5}	Fail
Gemma-2	-0.0451	0.1595	0.3134	1.0000	1.50×10^{-5}	Fail

17.2 Norm-, layer- and gate-matched empirical direction null

For each original held-out intervention, the structured local operator, precursor term, gate state, action weights and layer schedule were retained. At each intervention layer, a random vector was sampled from a 64-component PCA basis of training-group empirical state updates, projected orthogonally to the corresponding structured component and rescaled to its exact component norm. The 50 seeds therefore retained energy, location, timing and gate coverage while replacing only orientation; full-dimensional isotropic noise was not used. The endpoint is strict conflict-to-clean full-vocabulary top-1 crossings among closure prompts minus top-1 changes in non-closure prompts.

The nominal Gemma result does not survive correction across the three planned checkpoints. Thus the test does not licence a checkpoint-general direction-specificity claim. It does not erase the separately reported bounded boundary crossings under the declared gate/executor policy, whose null asks a different question about prompt-to-action assignment.

Supplementary Methods Table 10 | Empirical-subspace random-direction actuator null.

Specificity is the strict closure crossing count minus the non-closure full-vocabulary top-1 change count. The structured endpoint was independently reproduced before null sampling. BH correction is across the three checkpoints.

Checkpoint	Structured specificity	Null mean	Null 95th percentile	Empirical P	BH q	Family-adjusted direction-specificity verdict
Qwen	17	6.48	25.20	0.1373	0.1373	Fail
Llama	11	2.40	18.10	0.0980	0.1373	Fail
Gemma-2	9	0.02	1.55	0.0196	0.0588	Fail; nominal unadjusted result only

17.3 New-graph paired geometry confirmation

The 24 new graph groups used disjoint entity names, relation phrases and candidate labels from the original controlled set. Each group supplied the fixed ten condition types `clean`, `permuted`, `redundant`, `irrelevant`, `weak_distractor`, `competition_balanced`, `direct_conflict`, `closure_update`, `closure_override` and `exception_override`. The exact raw prompt string and prompt content digest were shared across the Qwen and Gemma runtimes. Geometry used all block outputs, with no functional window selection. Within each shuffle repeat one permutation of intermediate block indices was shared across all 240 prompts and endpoints were retained.

The Gemma checkpoint is a later-generation Gemma-3-12B instruction-tuned QAT Q4_0 GGUF checkpoint rather than a same-family controlled scale ablation. Its native capture was qualified by two byte-identical 48-block exports before the full run. The primary result only says that the measured all-layer path geometry recurs in these two checkpoint-specific settings.

Supplementary Methods Table 11 | Frozen paired new-graph chord-geometry confirmation.

Chord alignment is primary; detour and curvature are supporting metrics. Each empirical value is $1/51$, because the real mean exceeded all 50 shared endpoint-preserving shuffles. BH correction is across the two chord-alignment tests.

Checkpoint	Prompts	Blocks / hidden size	Real alignment	Null mean	Null s.d.	Null-standardized effect	Empirical P	BH q
Qwen2.5-1.5B-Instruct	240	28 / 1,536	0.07924	0.04180	0.00665	5.632	0.0196	0.0392
Gemma-3-12B-it-QAT-Q4_0	240	48 / 3,840	0.08640	0.02403	0.01839	3.392	0.0196	0.0392